

CASE STUDY – 1

Digital SME Lending & Repayment Platform

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Contents

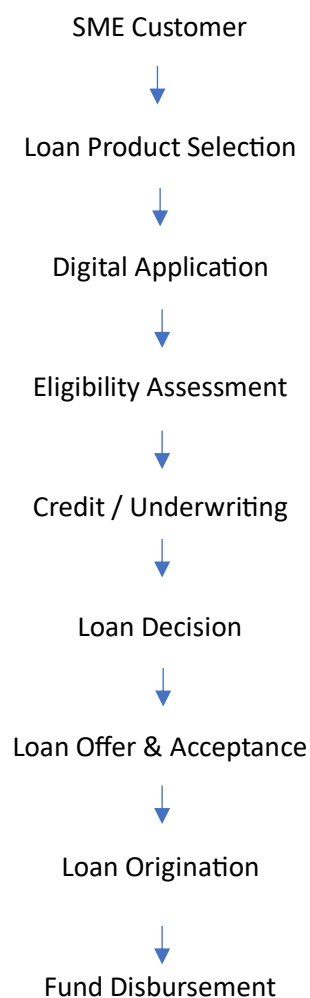
1. Project Snapshot:	3
1.1 High – level Business Flow	3
2. Project Summary:	4
3. Project Justification / Business Need:	5
4. Business Objectives:	5
5. Project Objectives:	5
Key Project Objectives	5
6. Business Rules:	6
7. Project Scope:	6
7.1 : In Scope:	6
7.2 : Out of Scope:	7
8. Stakeholder Analysis:	7
8.1 : RACI Matrix	7
8.2 : Stakeholder Analysis & Key Observations	8
9. As – Is Process Analysis:	8
Current – State:	8
10. Gap Analysis:	9
11. To – Be Process:	9
Future – State:	10
12. Key Requirements:	10
12.1 Functional Requirements:	10
12.2 Non - Functional Requirements:	11
13. User Stories with Acceptance Criteria:	11
14. BPMN / UML :	13
14.1 BPMN:	13
14.2 UML:	14
.....	14
15. Integration & API Analysis	14
15.1 Integration Overview	14
High-Level Integration Flow	14
15.2 Sample API Analysis	15
16. Data Mapping & SQL Analysis	15
16.1 Data Mapping	15

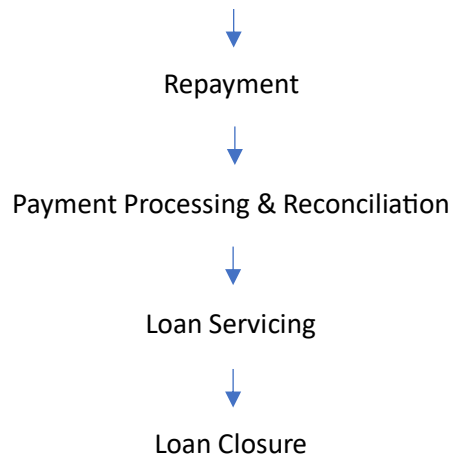
16.2 SQL Analysis.....	15
17. Dashboard & KPI Insights:	17
18. UAT:.....	17
19. Requirement Traceability Matrix (RTM):.....	18
20. Change Request & Impact analysis:.....	18
Impact Analysis.....	18
21. BA Recommendation / Expected Outcome:	19
Expected Outcomes	19

1. Project Snapshot:

Category	Details
Domain	Banking & Financial Services
Business Area	SME Lending
Target Users	SME Borrowers, Lending Teams, Credit & Underwriting Teams, Operations and Servicing Teams
Business Need	Streamline the SME lending journey by reducing manual activities, improving application visibility and enabling digital repayment management.
Core Capabilities	Digital Loan Application, Eligibility Assessment, Credit & underwriting Workflow, Loan Origination, Disbursement, Repayment Management, payment Tracking, Reconciliation and Loan Servicing
Primary BA Focus	Business Analysis, Requirements Management, Process Improvement, Business Rules, Agile Delivery, Integration Analysis, Data Analysis and UAT
Key Artifacts	BRD, As-Is/To-Be Process Models, Gap Analysis, Business Rules, User Stories, Acceptance Criteria, BPMN, UML, Data Mapping, API Analysis, SQL Analysis, Dashboard, UAT and RTM
Portfolio Type	Simulated Business Analysis Case study

1.1 High – level Business Flow





2. Project Summary:

Small and medium – sized businesses often require financing to support activities such as working capital, equipment purchases, business expansion, inventory requirements, and other operational needs.

In a traditional lending environment, the loan journey can involve multiple interactions between the borrower, relationship manager, credit team, underwriting team, operations, and servicing team. Information may need to be collected and validated at different stages before a lending decision can be made.

The proposed Digital SME Lending & Repayment Platform is designed to provide a centralised digital experience for the complete lending lifecycle.

The platform enables an SME customer to:

Discover a loan product -> Start an application -> Provide business information -> Submit supporting documents -> Receive an eligibility/preliminary decision -> Completed the lending process -> Receive funds -> Make repayments -> Track loan status.

For the lending organization, the platform provides capabilities for:

- Digital application intake
- Customer and business information management
- Loan eligibility assessment
- Credit and underwriting workflow
- Loan approval and offer management
- Loan disbursement
- Repayment collection
- Payment status tracking
- Payment reconciliation
- Loan servicing
- Operational reporting and dashboards

3. Project Justification / Business Need:

The existing SME Lending journey involves multiple manual interactions for collecting borrower information, reviewing applications, coordinating underwriting and managing repayments.

As application volumes increase, this can lead to:

- Longer processing time
- Repeated information collection
- Limited application visibility
- Manual coordination between teams
- Difficulty Tracking payment exceptions

Why the platform is Needed:

The organization wants to move from a manual, fragmented lending process to a centralized digital lending journey.

Business Need:

Reduce manual effort | Improve processing visibility | Standardize workflows | Simplify repayments
| Improve operational reporting |

4. Business Objectives:

The key business objectives of the Digital SME Lending & Repayment Platform are to:

1. **Reduce Manual Processing**
Minimize repetitive activities involved in SME loan application and servicing.
2. **Improve Processing Efficiency**
Enable a more streamlined journey from application through loan decision and servicing.
3. **Improve Customer Visibility**
Provide customers with visibility into application, loan, and repayment status.
4. **Standardize Lending Workflows**
Establish consistent processes across application, underwriting, origination, and servicing.
5. **Improve Repayment Management**
Provide better visibility into payment status and payment-related exceptions.
6. **Enable Data-Driven Decisions**
Provide reliable lending and repayment metrics through centralized reporting and dashboards.

5. Project Objectives:

Key Project Objectives

1. **Digital Loan Application**
Enable SMEs to submit loan applications and required information through a centralized platform.
2. **Automated Application Workflow**
Route applications through eligibility, assessment, underwriting, approval, and origination stages.

3. **Loan Status Tracking**
Allow customers and internal teams to track application and loan progress.
4. **Digital Repayment Management**
Enable repayment processing, payment-status tracking, and loan balance updates.
5. **Payment Reconciliation**
Identify and manage mismatches between loan records and payment transactions.
6. **Operational Reporting**
Provide dashboards and reports for monitoring applications, loans, and repayment performance.

6. Business Rules:

Rule ID	Business Rule
BR- 001	An SME loan application must contain all mandatory borrower, business, loan and supporting information before it can proceed for assessment.
BR - 002	The system shall allow only eligible loan products to be presented based on the information provided by the applicant.
BR - 003	Applications that do not meet the configured eligibility criteria shall not proceed to the next decision stage.
BR - 004	A loan cannot be originated until the required approval conditions have been satisfied.
BR - 005	The approved loan amount and terms shall be based on the applicable lending rules and approved decision.
BR - 006	A repayment transaction shall update the corresponding loan account only after the payment is confirmed as successful.
BR - 007	Duplicate payment transactions shall be identified and flagged for investigation.
BR – 008	Any mismatch between the expected repayment and received payment shall be recorded as a reconciliation exception.
BR – 009	Failed or reversed payments shall not be treated as successful repayments.
BR - 010	Changes to configured lending or pricing rules shall be restricted to authorized users and recorded for audit purposes.

7. Project Scope:

The project scope defines the capabilities included in the Digital SME Lending & Repayment Platform and the areas excluded from this case study.

7.1 : In Scope:

Area	Scope
Customer & Business Profile	Capture and maintain SME/customer information
Loan Application	Create, submit and manage SME loan applications
Eligibility Assessment	Validate application information against configured eligibility criteria
Credit & Underwriting	Support application assessment, review and underwriting workflow
Loan Decision	Support approval, rejection and conditional decision outcomes
Loan Origination	Create and manage approved loan records

Disbursement	Support the loan funding workflow
Repayment Management	Track scheduled and received loan repayments
Payment Tracking	Maintain payment status and transaction references
Reconciliation	Identify and manage payment mismatches and exceptions
Loan Servicing	Support ongoing loan account and repayment activities
Notifications	Send automated customer notifications for relevant application and payment events
Reporting & Dashboard	Provide operations metrics for applications, loans and repayments

7.2 : Out of Scope:

- Development of the core banking platform
- Development of external payment networks
- Creation of credit bureau services
- Direct implementation of banking infrastructure
- Regulatory reporting platform development
- Mobile application development
- Physical document processing

8. Stakeholder Analysis:

8.1 : RACI Matrix

R = Responsible | **A** = Accountable | **C** = Consulted | **I** = Informed |

Process / Activity	SME Borrower	Relationship Manger	Credit Analyst	Underwriter	Lending Operations	Servicing	Finance / Reconciliation	Product Owner	Technology
Customer Onboarding	R	A	C	I	R	I	I	C	C
Loan Application	R	A	C	I	R	I	I	C	C
Eligibility Assessment	C	C	R	C	R	I	I	C	C
Credit Assessment	I	C	R	A	C	I	I	I	I
Underwriting	I	C	C	A/R	C	I	I	I	I
Loan Approval	I	C	C	A/R	C	I	I	A	I
Loan Origination	I	C	C	C	A/R	I	C	I	C
Loan Disbursement	I	C	C	I	R	C	A	I	C

Repayment Processing	R	I	I	I	C	A/R	C	I	C
Payment Reconciliation	I	I	I	I	C	C	A/R	I	C
Loan Servicing	I	I	I	I	C	A/R	C	I	C
Reporting & Dashboard	I	C	C	C	R	C	C	A	R

8.2 : Stakeholder Analysis & Key Observations

The RACI analysis highlights the following key Observations:

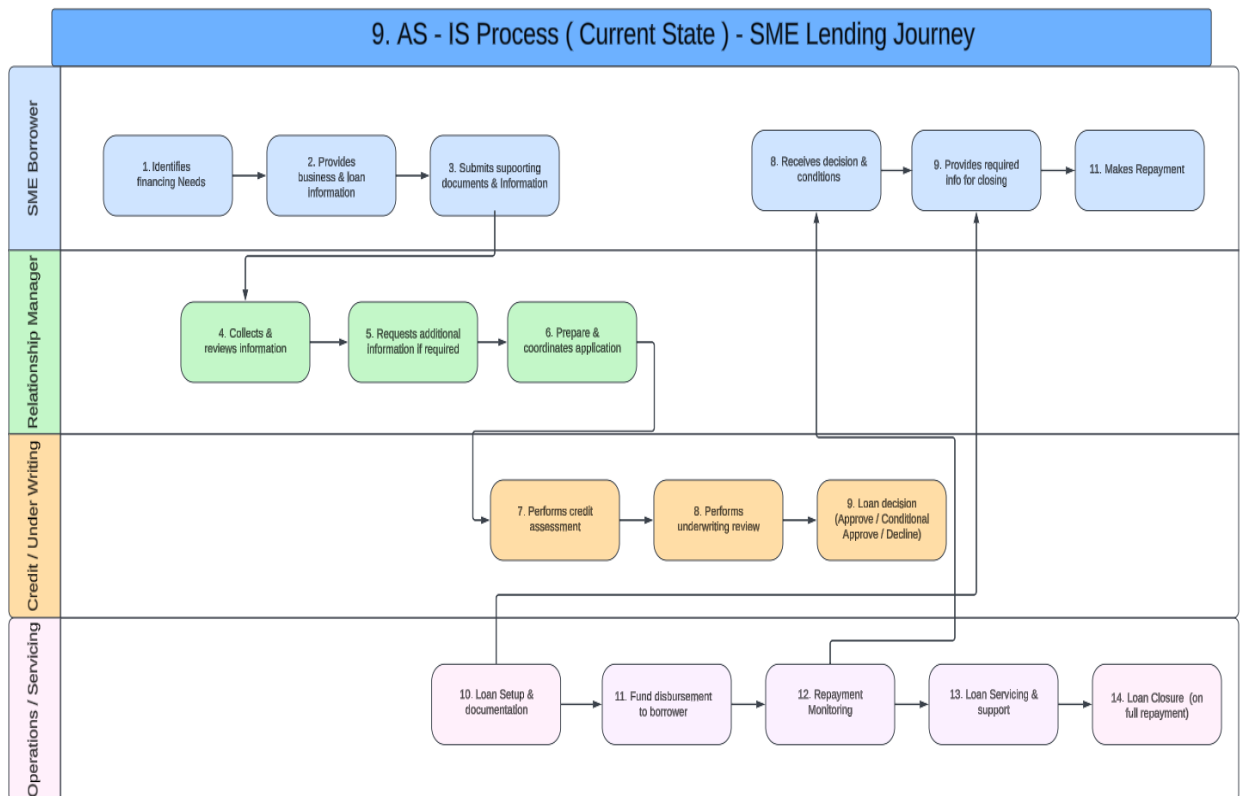
- The lending journey involves multiple cross – functional handoffs across Relationship Management, Credit, Underwriting, Operations, Finance and Servicing.
- Underwriting, loan origination, and payment reconciliation are key control points in the process.
- Lending Operations has significant responsibility across multiple stages, making it a potential area for process automation.
- Payment reconciliation requires coordination between Finance, Operations and Servicing.
- Clear ownership and workflow–based routing are important to minimise delays and information gaps.

9. As – Is Process Analysis:

Current – State:

The SME lending process is primarily dependent on manual interactions between the borrower, Relationship Manager, Credit / Underwriting, and Operations/ Servicing teams. The borrower provides business and loan information and supporting documents, while the Relationship Manager coordinates the application with Credit and Underwriting for assessment and decisioning. Once approved, Operations handles loan setup and disbursement, followed by repayment monitoring and servicing.

Swimlane Diagram of AS- IS State:



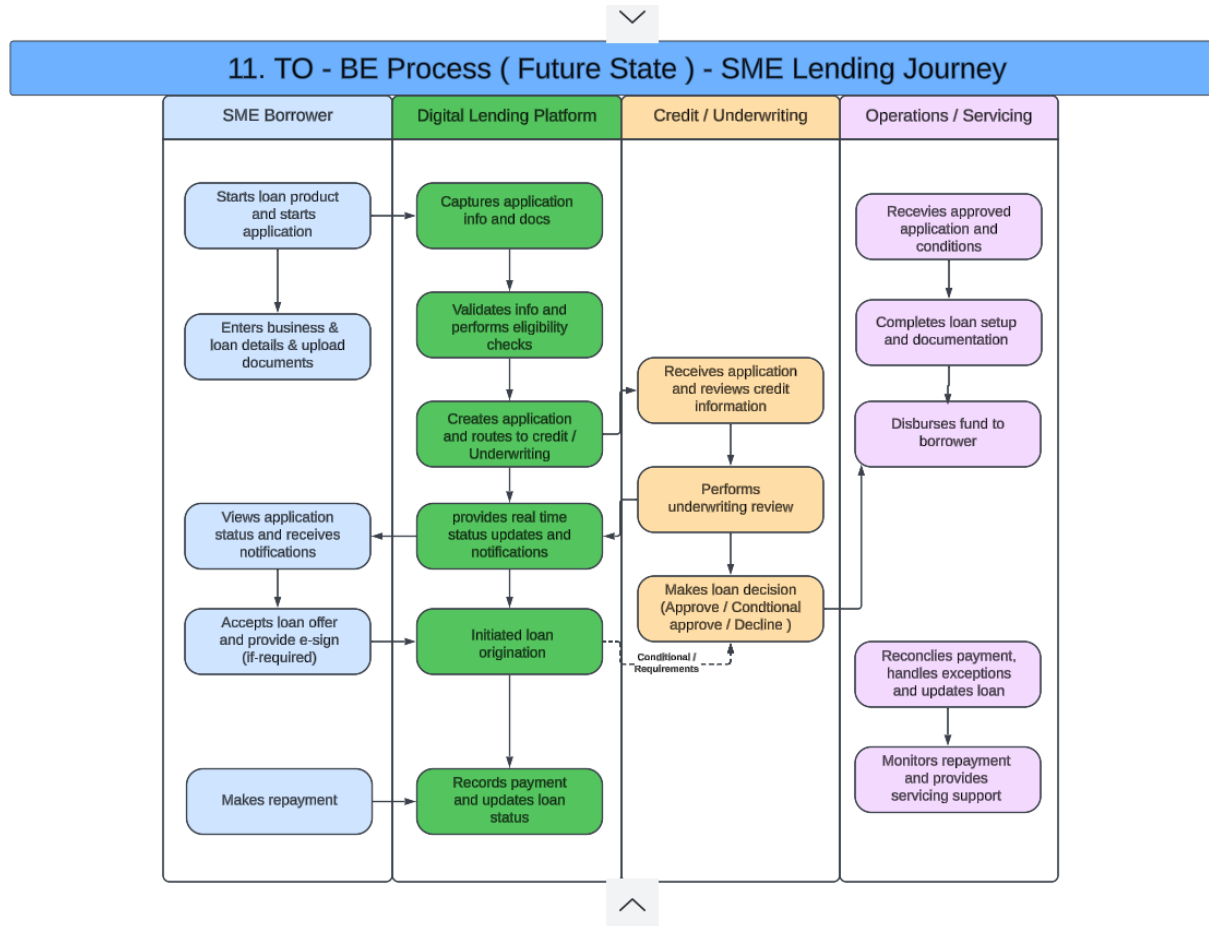
10. Gap Analysis:

- Manual application information collection -> Leads to repeated data collection and additional processing time.
- Multiple handoffs between lending teams -> Creates coordination delays and unclear ownership.
- Limited application status visibility -> Increases customer follow-ups and dependency on lending teams.
- Manual payment monitoring -> Creates additional operational effort in tracking repayments.
- Manual payment reconciliation -> Makes it harder to identify and resolve payment mismatches efficiently.
- Limited centralised reporting -> Makes it difficult for stakeholders to monitor lending and repayment performance.

11. To – Be Process:

Future – State:

The SME lending process will be digitised through a centralised, workflow-driven platform. Customers can submit applications, provide documents, and track progress digitally, while the platform validates information, performs eligibility checks, routes applications to credit/underwriting, supports loan origination, and enables digital repayment tracking, reconciliation, and servicing.



12. Key Requirements:

12.1 Functional Requirements:

S.No	Requirements
FR - 001	Customer Profile: The system shall allow SME customers to create and maintain their business and contact information
FR - 002	Loan Application: The system shall allow customers to select a loan product and submit a digital loan application
FR - 003	Document Submission: The system shall allow customers to upload required supporting documents as part of the application
FR - 004	Eligibility Validation: The system shall validate application information against configured eligibility criteria.

FR - 005	Application Workflow: The system shall route applications to the appropriate Credit and Underwriting teams based on workflow status.
FR - 006	Application Status: The system shall provide customers and internal users with the current application status
FR - 007	Loan Decision: The system shall support approval, conditional approval, and decline outcomes.
FR - 008	Loan Origination: The system shall create the loan record after required approval conditions are satisfied.
FR - 009	Repayment Management: The system shall record and track customer repayment transactions against the corresponding loan.
FR - 010	Payment Reconciliation: The system shall identify payment mismatches and create exceptions for investigation.
FR - 011	Notifications: The system shall send notifications for defined application, decision, and payment events.
FR - 012	Reporting: The system shall provide operational reporting for applications, loans and repayments.

12.2 Non - Functional Requirements:

- Security: Customer and financial information must be protected through appropriate access controls.
- Auditability: Important application, decision and payment changes must be traceable.
- Availability: The platform should be available to users during defined business operating periods.
- Performance: Key application screens and transactions should respond within agreed performance thresholds.
- Scalability: The solution should support increasing application and transaction volumes.

13. User Stories with Acceptance Criteria:

US – 001 – Submit Loan Application

User Story : As an SME borrower, I want to submit my loan application digitally so that I can provide the required information without repeated manual interactions.

Acceptance Criteria:

- The borrower can enter all mandatory application details.
- The system validates mandatory fields before submission.
- A unique application ID is generated after successful submission.
- The borrower can view the submitted application status.

US – 002 – Eligibility Assessment

User Story: As a lending user, I want the system to validate application information against configured eligibility criteria so that only eligible applications proceed.

Acceptance Criteria:

- The system evaluates the required eligibility criteria.
- Eligible applications proceed to the next stage.
- Ineligible applications are stopped or routed according to the configured workflow.
- The eligibility outcome is recorded against the application.

US – 003 – Application Status Tracking

User Story: As an SME borrower, I want to view my application status so that I know where my application is in the lending process.

Acceptance Criteria:

- The system displays the current application status.
- Status changes are reflected in the customer view.
- The borrower receives notifications for defined status changes.

US – 004 – Loan Decision

User Story: As an underwriter, I want to record an approval, conditional approval, or decline decision so that the application can proceed according to the outcome.

Acceptance Criteria:

- The underwriter can select the applicable decision.
- Conditional approval allows required conditions to be recorded.
- The decision is stored against the application.
- The appropriate users/customer are notified.

US – 005 – Make Loan Repayment

User Story: As a borrower, I want to make my loan repayment digitally so that I can manage my repayment without manual follow-up.

Acceptance Criteria:

- The borrower can view the amount due.
- A repayment can be initiated for an active loan.
- A successful payment is recorded against the loan.
- The loan balance/payment status is updated after confirmation.

US – 006 – Payment Reconciliation Exception

User Story: As a finance/operations user, I want payment mismatches to be identified so that discrepancies can be investigated and resolved.

Acceptance Criteria:

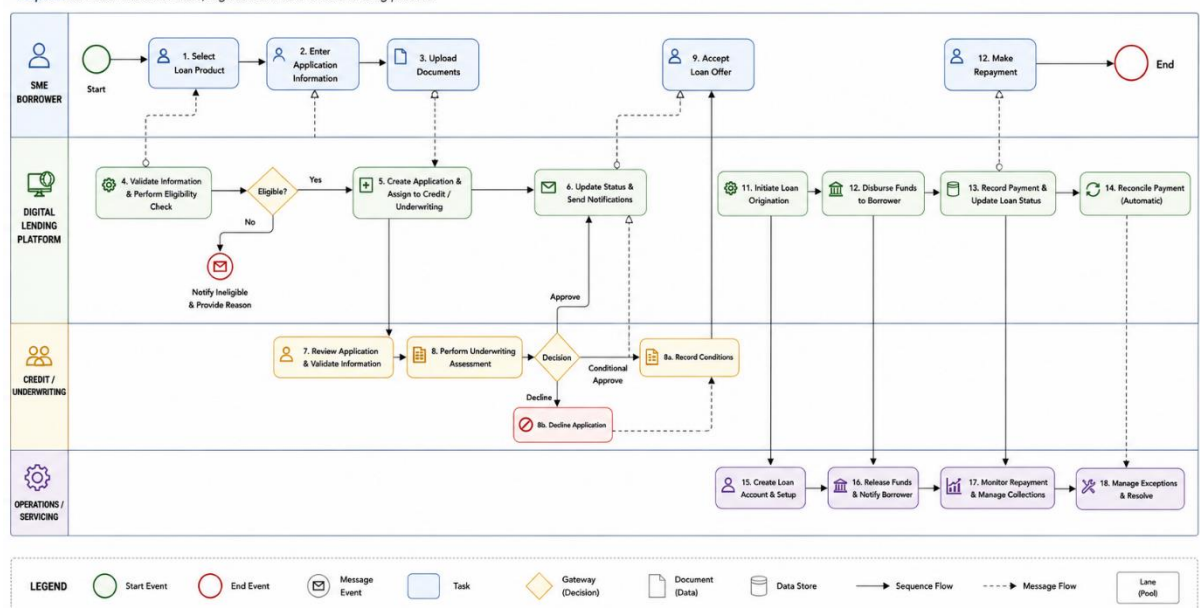
- The system compares expected and received payment information.
- A mismatch is flagged as an exception.
- The exception contains sufficient transaction/reference information for investigation.
- The exception remains open until resolved.

14. BPMN / UML :

14.1 BPMN 2.0 - To Be Lending Process:

BPMN 2.0 – To-Be SME Lending Process

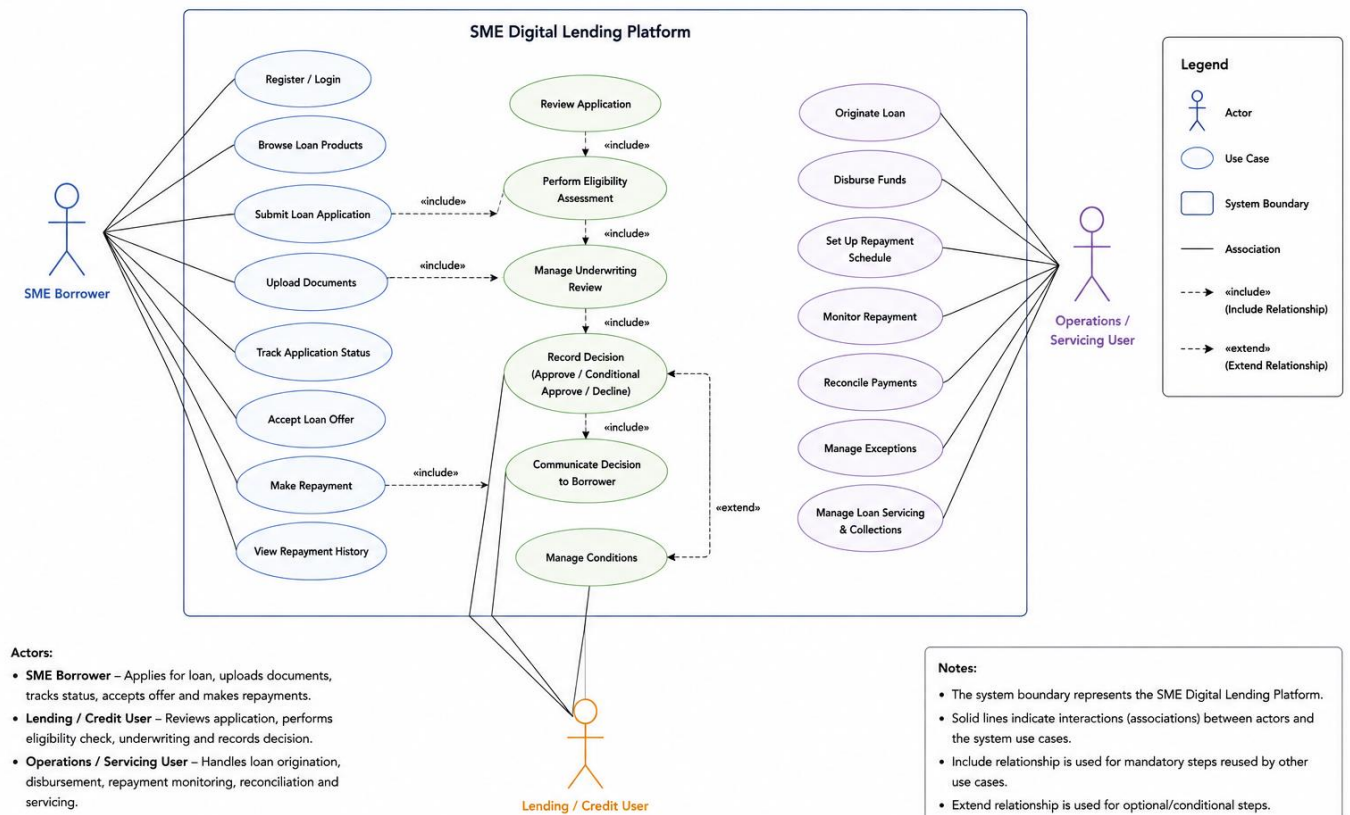
Purpose: To model the future state, digitized end-to-end SME lending process.



14.2 UML Use Case Diagram:

UML Use Case Diagram – SME Digital Lending Platform

This diagram shows the high-level interactions between different users (actors) and the SME lending system.



15. Integration & API Analysis

15.1 Integration Overview

The Digital SME Lending & Repayment Platform interacts with supporting services such as credit assessment, core banking, payment processing, and notification services. The platform exchanges application, decision, loan, and payment information with these systems to support the end-to-end lending lifecycle.

High-Level Integration Flow

Digital Lending Platform



Credit / Eligibility Service



Core Banking / Loan Account



Payment Service



Notification Service

15.2 Sample API Analysis

API / Integration	Purpose	Key Data
Credit / Eligibility API	Validate borrower/application eligibility	Customer ID, application details, eligibility result
Loan Origination API	Create approved loan record	Application ID, approved amount, loan terms
Disbursement API	Initiate fund disbursement	Loan ID, amount, destination account
Payment API	Initiate / track repayment	Loan ID, payment amount, payment reference, status
Notification API	Send customer notifications	Customer ID, event type, message

16. Data Mapping & SQL Analysis

16.1 Data Mapping

Source System	Source Field	Target System	Target Field
Lending Platform	Application ID	Loan System	Loan Application ID
Lending Platform	Customer ID	Loan System	Customer ID
Lending Platform	Approved Amount	Loan System	Loan Amount
Payment Service	Transaction ID	Lending Platform	Payment Reference
Payment Service	Payment Amount	Loan System	Repayment Amount
Payment Service	Payment Status	Lending Platform	Payment Status

16.2 SQL Analysis

Business Question 1: What is the payment success rate?

SELECT

ROUND(

SUM(CASE WHEN payment_status = 'SUCCESS' THEN 1 ELSE 0 END)

* 100.0 / COUNT(*), 2

) AS payment_success_rate

FROM payments;

Illustrative Result: 80%

Insight: 20% of payment transactions were unsuccessful and may require operational follow-up.

Business Question 2: Which applications are still pending?

SELECT

loan_product,

SUM(approved_amount) AS total_approved_amount

FROM loan_applications

WHERE application_status = 'APPROVED'

GROUP BY loan_product

ORDER BY total_approved_amount DESC;

Loan Product	Approved Amount
Working Capital	\$4.8M
Equipment Finance	\$3.2M
Business Expansion	\$2.6M

Insight: Working Capital has the highest approved lending value in the illustrative dataset.

Business Question 3: What is the total outstanding loan amount by product?

SELECT

payment_id,

loan_id,

payment_amount,

expected_amount

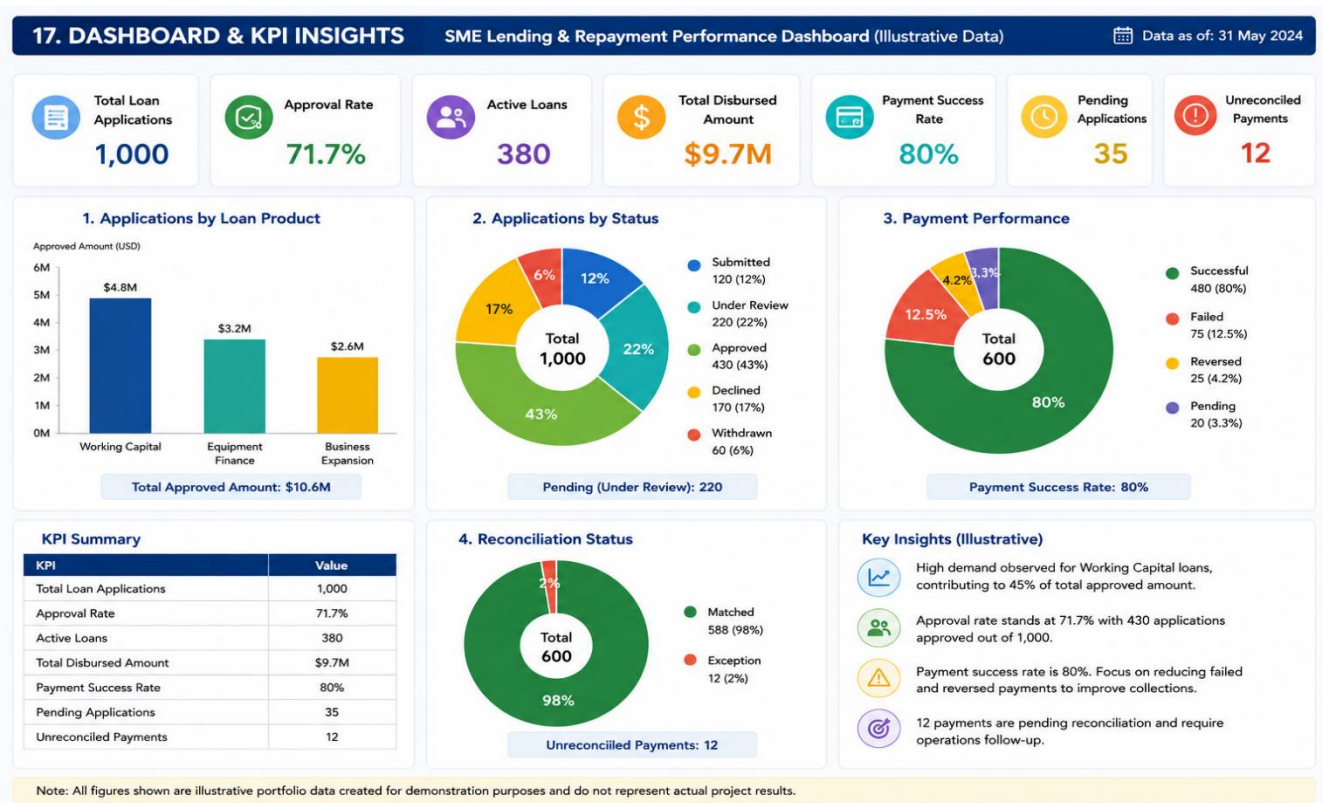
FROM payments

WHERE payment_amount <> expected_amount;

Illustrative Result: 12 transactions

Note: SQL queries and results are based on illustrative portfolio data created to demonstrate Business Analysis and data-analysis capability.

17. Dashboard & KPI Insights:



19. Requirement Traceability Matrix (RTM):

Business Requirement	Functional Requirement	User Story	UAT
Digital loan application	FR-002 – Submit Loan Application	US-001	UAT-001
Application eligibility validation	FR-004 – Eligibility Validation	US-002	UAT-003
Application status visibility	FR-006 – Application Status	US-003	UAT-001
Loan decision management	FR-007 – Loan Decision	US-004	UAT-004
Digital repayment	FR-009 – Repayment Management	US-005	UAT-005
Payment reconciliation	FR-010 – Payment Reconciliation	US-006	UAT-006

20. Change Request & Impact analysis:

CR-001: Introduce an additional digital repayment method for SME borrowers.

The business wants to provide an additional repayment option to improve customer convenience and reduce dependency on the existing payment method.

Impact Analysis

Area	Impact	Priority
Customer Payment UI	New repayment option required	High
Payment Integration	New payment processing flow	High
Loan Account	Payment status/balance update	High
Reconciliation	New transaction type must be matched	High
Notifications	Payment confirmation/failure messages	Medium
UAT	New positive and exception scenarios	High
Reporting	Payment metrics may need updating	Medium

21. BA Recommendation / Expected Outcome:

The proposed Digital SME Lending & Repayment Platform provides a centralized and streamlined lending journey by reducing manual interactions, improving application visibility, supporting digital repayments, and strengthening payment reconciliation. The solution is expected to improve operational efficiency, enhance the customer experience, and provide better visibility into lending and repayment performance.

Expected Outcomes

- Reduced manual effort across the lending lifecycle
- Improved application and loan-status visibility
- Faster and more consistent processing
- Improved repayment tracking and reconciliation
- Better operational reporting and decision-making

Note: Outcomes are expected benefits for this simulated portfolio case study and are not actual project results.